AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH (AIUB)

Faculty of Science and Technology (FST)
Department of Computer Science (CS)
Undergraduate Program

COURSE PLAN	SEMESTER: Summer 2024-2025
I. Course Code and Title: CSC 3113 Theory of Computation **II. Credit**: 3 Hour Theory per week **III. Nature** Core Course for CSE **IV. Prerequisite** CSC 2211 Algorithms	**V. Vision:** Our vision is to be the preeminent Department of Computer Science through creating recognized professionals who will provide innovative solutions by leveraging contemporary research methods and development techniques of computing that is in line with the national and global context. **VI. Mission:** The mission of the Department of Computer Science of AIUB is to educate students in a student-centric dynamic learning environment; to provide advanced facilities for conducting innovative research and development to meet the challenges of the modern era of computing, and to motivate them towards a life-long learning process.

VII - Course Description:

- Understanding the notations used in computer science literature.
- Understanding the mathematical model of Computation.
- Use of Computational models to solve problems.
- Understanding Computability
- Determining Complexity of problems

VIII - Course outcomes (CO) Matrix:

By the end of this course, students should be able to:

COs*	CO Description	Level of Domain***			PO Assessed ****
		C	P	A	
CO1	**Demonstrate** different computational model and mathematical notations	3			PO-a-2
CO2	**Use** original example of different computational model and mathematical notations	3			PO-a-2
CO3	**Illustrate** a solution for a complex problem using the principles of existing computational models	4			PO-c-3
CO4 **	**Explain** the principles of existing computational models and use it to find a new solution.			4	PO-l-2

C: Cognitive; P: Psychomotor; A: Affective Domain

* *CO assessment method and rubric of COs assessment is provided in later section*
** *COs will be mapped with the Program Outcomes (POs) for PO attainment*
*** *The numbers under the 'Level of Domain' columns represent the level of Bloom's Taxonomy each CO corresponds to.*
**** *The numbers under 'PO Assessed' column represent the POs each CO corresponds to.*

IX - Topics to be covered in the class and/or lab *

Time Frame	CO Mapped	Topics	Teaching Activities	Assessment Strategy(s)
Week 1	CO1, CO2	Mission & Vision of AIUB, OBE Guidelines and assessment policies, Basic Mathematical Concepts Finite Automaton, Deterministic Finite Automaton (DFA)	Discussion on Mission & Vision of AIUB, Introduction to Theory of Computation Review of Pre-requisite study materials, perform of exercises	
Week 2	CO3, CO4	DFA	Discussion, Group study and perform of exercises, PPT slides, board work.	Group study, Homework, Quiz
Week 3	CO4	Non-determinism and non-regular languages	Discussion, Group study and perform of exercises, PPT slides, board work.	
Week 4	CO1, CO2	Regular Expression	Discussion, Group study and perform of exercises	Pop Quiz, Class discussion, question answer session, Homework
Week 5	CO1, CO2	Regular Expression	Discussion, Group study and perform of exercises	Pop Quiz, Class discussion, question answer session, Homework
Midterm (Week 6)				
Week 7		Context Free Grammer	Discussion, Group study and perform of exercises, PPT slides, board work	Pop Quiz, Class discussion, question answer session
Week 8		Ambiguity, Chomsky Normal Form	Discussion, Group study and perform of exercises, PPT slides, board work	Pop Quiz, Class discussion, question answer session
Week 9		Push Down Automata	Discussion, Group study and perform of exercises, PPT slides, board work	Pop Quiz, Class discussion, question answer session
Week 10		Turing Machine	Discussion, Group study and perform of exercises, PPT slides, board work	Pop Quiz, Class discussion, question answer session
Week 11		Turing Machine	Discussion, Group study and perform of exercises, PPT slides, board work	Quiz, Class discussion, question answer session
Final term (Week 12)				

* *The faculty reserves the right to change, amend, add, or delete any of the contents.*

X - Mapping of PO to Courses and K, P, A

PO Indicator ID	PO Indicators Definition (As per the requirement of WKs)	Domain	K	P	A
PO-a-2	Apply information and concepts of *mathematics* with the familiarity of issues.	Cognitive Level 3 (Applying)	K2		

PO-c-3	Develop computer science and engineering solutions that meet specified needs with appropriate environmental considerations.	Cognitive Level 4 (Analyzing)	K5	P1 P2 P6	
PO-l-2	Seek and use resources in solving computer science and engineering problems.	Affective Level 4 (Organizing)			

XI – K, P, A Definitions

Indicator	Title	Description
K2	Conceptual based mathematics	Conceptually based mathematics, numerical analysis, statistics and the formal aspects of computer and information science to support analysis and modeling applicable to the discipline
K5	Engineering Design	Knowledge that supports engineering design in a practice area
P1	Depth of knowledge required	Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6 or K8 which allows a fundamentals-based, first principles analytical approach
P2	Range of conflicting requirements	Involve wide-ranging or conflicting technical, engineering, and other issues
P6	Extent of stakeholder involvement and conflicting requirements	Involve diverse groups of stakeholders with widely varying needs

XII – Mapping of CO Assessment Method and Rubric

The mapping between Course Outcome(s) (COs) and The Selected Assessment method(s) and the mapping between Assessment method(s) and Evaluation Rubric(s) is shown below:

COs	Description	Mapped POs	Assessment Method	Assessment Rubric
CO1	Demonstrate different computational model and mathematical notations	PO-a-2	Quiz	Rubric for Quiz
CO2	Use original example of different computational model and mathematical notations	PO-a-2	Quiz	Rubric for Quiz
CO3	Illustrate a solution for a complex problem using the principles of existing computational models	PO-c-3	Quiz	Rubric for Term exam
CO4	Explain the principles of existing computational model and use it to find a new solution.	PO-l-2	Midterm Exam	Rubric for Term exam

XIII – Evaluation and Assessment Criteria

CO1 [PO-a2]: Demonstrate different computational model and mathematical notations						
Assessment Attribute/ Criteria	**Missing (0)**	**Inadequate (1)**	**Not enough (2)**	**Moderate (3)**	**Satisfactory (4)**	**Excellent (5)**

Definition	Not Attended/ Incorrect	While providing the definitions, the student Described one Element	While providing the definitions, the student Described two Element	While providing the definitions, the student Described three Element	While providing the definitions, the student Described four Element	While providing the definitions, the student Described five Element
Logical Argument	Not Attended/ Incorrect	Described one Elements	Described two Elements	Described three Elements	Described four Elements	Described five Elements
Relevant Example	Not Attended/ Incorrect	Showed one element in the example	Showed two elements in the example	Showed three elements in the example	Showed four elements in the example	Showed five elements in the example

CO2 [PO-a2]: Use original example of different computational model and mathematical notations

Assessment Attribute/ Criteria	**Missing (0)**	**Inadequate (1)**	**Not enough (2)**	**Moderate (3)**	**Satisfactory (4)**	**Excellent (5)**
Definition	Not Attended/ Incorrect	While providing the example, the student Described one Element	While providing the example, the student Described two Element	While providing the example, the student Described three Element	While providing the example, the student Described four Element	While providing the example, the student Described five Element
Logical Argument	Not Attended/ Incorrect	Described one Elements	Described two Elements	Described three Elements	Described four Elements	Described five Elements
Relevant Example	Not Attended/ Incorrect	Showed one element in the example	Showed two elements in the example	Showed three elements in the example	Showed four elements in the example	Showed five elements in the example

CO3 [PO-c-3]: Illustrate a solution for a complex problem using the principles of existing computational models

Assessment Attribute/ Criteria	**Missing (0)**	**Inadequate (1)**	**Not enough (2)**	**Moderate (3)**	**Satisfactory (4)**	**Excellent (5)**

Design	Not Attended/ Incorrect	While designing the solution, the student used one element correctly	While designing the solution, the student used two elements correctly	While designing the solution, the student used three elements correctly	While designing the solution, the student used four elements correctly	While designing the solution, the student used five elements correctly
Design Conventions	Not Attended/ Incorrect	While designing the solution, the student followed the convention and used one element correctly	While designing the solution, the student followed the convention and used two elements correctly	While designing the solution, the student followed the convention and used three elements correctly	While designing the solution, the student followed the convention and used four elements correctly	While designing the solution, the student followed the convention and used five elements correctly
Result Analysis	Not Attended/ Incorrect	While analyzing the solution, the student identified the initial setting	While analyzing the solution, the student knows the formal definition of computation	While analyzing the solution, the student applied the formal definition of computation	While analyzing the solution, the student reached the final setup	While analyzing the solution, the student made the final decision

CO4 [PO-I-4]: Explain the principles of existing computational model and use it to find a new solution.

Assessment Attribute/ Criteria	**Missing (0)**	**Inadequate (1)**	**Not enough (2)**	**Moderate (3)**	**Satisfactory (4)**	**Excellent (5)**
Modification	Not Attended/ Incorrect	While modification the solution, the student calculated one Element correctly	While modification the solution, the student calculated two elements correctly	While modification the solution, the student calculated three elements correctly	While modification the solution, the student calculated four elements correctly	While modification the solution, the student calculated five elements correctly
Conventions Followed	Not Attended/ Incorrect	While modification, the student followed the convention to calculate one element correctly	While modification, the student followed the convention to calculate two elements correctly	While modification, the student followed the convention to calculate three elements correctly.	While modification, the student followed the convention to calculate four elements correctly.	While modification, the student followed the convention to calculate five elements correctly.
Result Analysis	Not Attended/ Incorrect	While analyzing the solution, the student identified the initial state	While analyzing the solution, the student knows the formal procedure	While analyzing the solution, the student applied the formal procedure	While analyzing the solution, the student reached the final setup	While analyzing the solution, the student made the final decision

XIV- Course Requirements

- Students are expected to attend at least 80% of the class.
- Students are expected to participate actively in the class.
- For both terms, there will be at least 2 quizzes based on the theoretical knowledge and conceptual understanding of the topic covered discussed in the classes.
- Submit report based on the given course related problems.
- Submission of assignments and projects should be in due time.

XV – Evaluation & Grading System*

The following grading system will be strictly followed in this class.

MID TERM		FINAL TERM	
Attendance	10%	Attendance	10%
Quiz	30%	Quiz	30%
Midterm written exam	60%	Final term written exam	60%
Total	100%	Total	100%
Grand Total 100% = 40% of Midterm + 60% of Final Term			

Letter	Grade Point	Numerical %
A+	4.00	90-100
A	3.75	85 - < 90
B+	3.50	80 - < 85
B	3.25	75 - < 80
C+	3.00	70 - < 75
C	2.75	65 - < 70
D+	2.50	60 - < 65
D	2.25	50 - < 60
F	0.00	< 50
I		Incomplete
W		Withdrawal
UW		Unofficially Withdrawal

** The evaluation system will be strictly followed as par with the AIUB grading policy.*
** CO attainment will be achieved with 60% of the evaluation marks.*

XVI – Textbook/ References

1. Introduction to the Theory of Computation (Latest Edition)
   by Michael Sipser (Textbook)
2. Introduction to Automata Theory, Languages, and Computation (Latest Edition)
   by John E. Hopcroft, et al
3. Elements of the Theory of Computation (Latest Edition)
   by Harry R. Lewis, Christos H. Papadimitriou